

FIT3080 Applied Class on Intelligent Agents**Exercise 1: Rationality**

Describe the requirements of a rational vacuum agent for the case where each movement costs one point, assuming that once a square has been cleaned, it doesn't get dirty.

Exercise 2: PEAS

For each of the following activities, give a PEAS description of the task environment, and characterize the task environment in terms of the environment types presented in the lecture slides.

- (a) Playing soccer.
- (b) Performing a high jump.
- (c) Knitting a sweater.

Exercise 3: Agent types

For each task in Exercise 2, consider the following types of agents, and for each type, describe what the agent can do, and how it differs from simpler agents: Simple reflex, Model based, Goal based and Utility based.

In addition, for each task, define *one* performance measure for which all the agents would be rational. Justify your answers.

Exercise 4: Vacuum-cleaner World and Agent (Python code)

- (a) Install a Python environment on your device. You can use Anaconda, Jupyter Notebook, or any other Python IDE of your choice.
- (b) Define Percepts and Actions using a suitable data structure in Python:
 - Percepts: Represent the state of the vacuum cleaner's world, e.g., ["A", "Dirty"] or ["B", "Clean"].
 - Actions: Define possible actions the vacuum cleaner can take, e.g., "Left", "Right", "Vacuum".
- (c) Write a program that implements a rational agent in the Vacuum-cleaner World. Your program should:
 - Read the current percept (location and state) of the vacuum cleaner.
 - Determine the appropriate action to take based on the current percept.
 - Execute the action and update the percept accordingly.